Beyond the Blob: Localizing Inserted Lesions with Aligned Vessel Forecasts

Anonymous CVPR submission

Abstract

Blood vessels are the main reason why nodule detectors in chest CT look in the wrong place: where a vessel bends or branches it produces a bright, compact spot that resembles a small lesion. Predicting the vessels from the surrounding tissue and removing them before the search is an appealing remedy, but it has been unclear whether such a vessel forecast helps because it is in the right place or merely because it gives the detector more information. We answer this question with a controlled experiment in which each lesion-free image region and its copy with an inserted lesion share the same surroundings, and therefore the same vessel forecast, while a rotated copy of the forecast serves as a misplaced control. The answer is that placement matters, but mainly for where the detector looks rather than for how many regions it accepts. At the same false-alarm rate, a correctly placed forecast moves the detector's mark onto the lesion for about 6 additional insertions per 100 on held-out LUNA16 scans, and by about 3 per 100 on scans with expert-annotated native vessels, whereas the number of accepted regions barely changes. Simply subtracting the forecast works as well as more flexible models of the vessel. These results suggest that vessel-aware detectors should be evaluated by where they point, not only by what they accept.

1. Introduction

Candidate scorers for pulmonary nodule detection respond to compact, approximately spherical bright structures in chest CT. Many such responses are produced by vessels: a vessel imaged in cross-section, a bend, or a junction all yield compact peaks (Fig. 1a). A scorer must therefore make two related decisions, namely whether a candidate crop contains a lesion and where within the crop the lesion lies. Vessels interfere with both. They can raise the score of a lesion-free crop, and within a crop that does contain a lesion they can produce the maximum of the spatial search, so that the location presented to a reader falls on the vessel.

Because vessels that traverse a candidate region are usually visible where they enter and leave it, their course through the region can be forecast from the surrounding tissue and accounted for before the search (Fig. 1c). Demonstrating that such a forecast helps for the intended reason is less straightforward than it may appear. A forecast adds input and model capacity, and it correlates with the type of background in the crop; each of these effects could raise a score even if the forecast were spatially misplaced.

In this work we test a specific hypothesis: a vessel forecast improves candidate scoring *because it is aligned with the anatomy*. The experimental design rests on two controls. First, each control crop and its lesion-bearing counterpart are identical outside a 14-mm ball that contains the lesion, so a forecaster that observes only the exterior produces the same forecast for both (Fig. 1b). Second, a sham forecast is obtained by rotating the same field by a quarter-turn about the slice axis; the sham retains every value, scale and degree of smoothness of the forecast but no longer coincides with the vessels (Fig. 1c). A third control supplies the classifier with summary statistics of the exterior only. We further examine how much flexibility the vessel estimate requires by evaluating a single local scoring function, which we call a *readout*, under four treatments of the vessel amplitude ranging from no forecast to a joint fit with the lesion (Sec. 3.3).

To see why placement can change the returned location without changing the decision, consider the synthetic slice of Fig. 1. Normalized to the vessel junction, the compact-blob score is 1.00 at the junction and 0.91 at the lesion, so the search returns the junction. Subtracting the aligned forecast lowers the junction score to 0.00 while the lesion keeps 0.89: the maximum over the crop falls by only about a tenth, so a crop accepted at a threshold below 0.89 stays accepted, but the returned location moves to the lesion. Subtracting the rotated forecast leaves the junction as the maximum (1.50 against 1.08 at the lesion), because the misplaced field removes nothing from the vessel and its subtraction even sharpens the contrast around it. More generally, if conditioning lowers the vessel and lesion scores by d_v and d_ℓ, the returned location moves to the lesion exactly when $d_v - d_\ell > m(x_v) - m(x_\ell)$, regardless of whether the maximum crosses the acceptance threshold. Our experiments show that this mechanism, rather than a change in acceptance, is what alignment provides on held-out data.

Our paper offers the following contributions. First, we

introduce a paired placement test that separates the effect of forecast placement from the information a forecast carries, evaluated at a fixed candidate false-positive budget. Second, within this design, a comparison of readouts under a shared search and classifier shows that subtracting the forecast at a fixed amplitude performs as well as estimating the amplitude jointly with the lesion, and we explain this with a short analysis of how a gain fitted inside the search region absorbs part of the lesion. Third, a localization analysis on a held-out LUNA16 set and on an independent dataset with annotated native vessels shows that alignment mainly changes the location returned by the scorer rather than the number of crops it accepts, which is the consequence of vessel forecasting that matters when a reader inspects the returned location.

2. Related Work

Vascular structure has long been treated as the main source of false compact responses in pulmonary nodule detection. Multiscale Hessian filters characterize local curvature and separate tubular from compact structures [5, 9, 12], but junctions produce curvature along several axes, so the strongest response in a search region may still lie away from the lesion. Agam *et al*. [1] therefore used reconstructed vessel trees to resolve such local ambiguities in CT, and Charbonnier *et al*. [3] used vessels outside a subsolid nodule to guide vessel segmentation within it. Learned candidate systems exploit local anatomy or multiple views [14, 16], and Gu *et al*. [7] added a vessel-suppression branch to a nodule detector. These approaches establish that vascular context is informative, but they change the input, the model and the context together, and none isolates whether the benefit comes from the spatial correspondence between the vessel estimate and the anatomy. From a signal-processing perspective, our readouts address the classical problem of detecting a nonnegative signal in the presence of structured interference [13], with a weighted affine background and a single exterior-derived nuisance component; what we vary is the origin and placement of that component and the treatment of its amplitude.

Our evaluation builds on two further lines of work. Inserting simulated lesions into patient CT produces hybrid images in which the location, size and morphology of the target are known; Robins *et al*. [11] compared projection- and image-domain techniques for virtual lung nodule insertion, and Gong *et al*. [6] proposed learned lesion and noise insertion for virtual clinical trials in chest CT. We use image-domain insertion not to emulate clinical prevalence but as a paired intervention, in which control and lesion crops share every exterior voxel so that the forecast is fixed and a placement sham becomes possible. Task-based assessment, in turn, has shown that pixel-level fidelity need not translate into better detection: Li *et al*. [8] compared denoisers across detection observers, and Yu *et al*. [17] reported that image-quality measures and detection efficiency can move in opposite directions. Following this view, we judge vessel subtraction by sensitivity at a fixed false-positive budget and by whether the returned location coincides with the inserted target, rather than by how well the vessel is removed.

3. Method

3.1. Problem Setup and Paired Insertions

Each input is a CT crop y of 64^3 voxels with spacing $1.5 \times 0.7 \times 0.7\,\mathrm{mm}$ (slice, row, column), centered on a candidate proposal c. We distinguish three nested regions: the search *core* within $10\,\mathrm{mm}$ of c, the *hidden ball* within $14\,\mathrm{mm}$, and the *exterior* E beyond $14\,\mathrm{mm}$, whose inner part between 15.5 and $19.5\,\mathrm{mm}$ we call the *shell*. A scorer consists of a readout, which maps the crop to template scores, and a *classifier* on top of them. It produces a crop decision $A(y) \in \{0, 1\}$, which accepts the crop for review, and a location $L(y)$, the single point it returns.

Let b denote a CT background, v an optional added vessel field, and s an inserted lesion with $s = 0$ on E. The control and lesion crops are

$$y_0 = b + v, \qquad y_1 = b + v + s. \tag{1}$$

For native-vessel backgrounds we set $v = 0$, since the vessels are already part of b. Because y_0 and y_1 coincide on E, the forecast $f = F(y|_E)$ is identical for a control and all of its insertions. All scorers evaluate y_0 and y_1 at the same center c and may use the entire crop; only the forecaster is restricted to E. A score can thus increase because a lesion is present, but not because the lesion altered the forecast.

Placement sham and exterior-summary control. Let Q denote a $90°$ rotation about the slice axis through c. Since the row and column spacings are equal, Q permutes voxels without interpolation. The sham forecast Qf preserves the radial profile, the multiset of values, the smoothness and the shape of f, but in general moves its vessels to locations without vascular anatomy. We refer to f and Qf as the *aligned* and *rotated* placements. Each aligned scorer is compared with (i) the rotated placement with a refitted classifier, which allows the sham to make the best possible use of a misplaced field, and (ii) the rotated placement evaluated with the aligned classifier, which removes any effect of refitting. A third control, denoted CT + shell, receives four summary statistics of the shell but no spatial field. Gains over all three controls are attributed to placement.

3.2. Exterior Vessel Forecast

The forecaster continues vessels whose entry and exit points are visible in the exterior. Intensity peaks on a sphere of radius R_a just outside the hidden ball propose pairs of entry and exit points, and a pair is retained if the signal continues

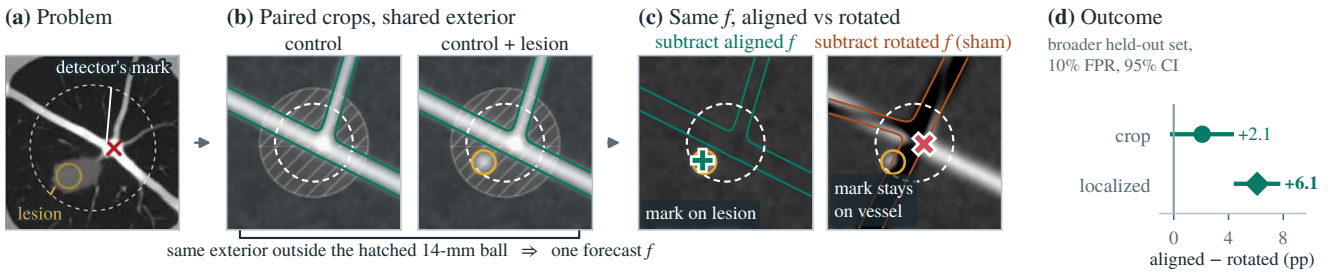


Figure 1. Problem, design and outcome. (a) A compact-blob search in the 10-mm core (dashed) returns the vessel junction (red cross) instead of the lesion (gold ring). (b) A control and its lesion copy share the exterior outside the hatched 14-mm ball, so both receive the same forecast f. (c) Subtracting the aligned f moves the returned location to the lesion; the rotated sham leaves it on the vessel. (d) On the broader held-out set, alignment changes crop sensitivity only slightly but raises localized sensitivity (95% CIs). Panel (a) was created in BioRender; (b) and (c) use a generated slice whose marks are computed with a simplified readout.

against its transverse flanks at two larger radii $R_1 < R_2$ on both sides. For each retained chord, a straight segment through the hidden ball, we fit a Gaussian cross-section with width and gain to the two exterior segments, interpolate width and amplitude along the chord, and taper the field smoothly beyond a radius R_t. The resulting forecast f is an additive estimate in CT units; a crop without an accepted chord has $f = 0$ and is processed by the same procedure. We verify the shared-exterior property by confirming that the rotation preserves the value multiset of every control, that rerunning the forecaster after replacing the hidden region yields an identical field, and that the visible exteriors of every rendered pair agree bitwise. A straight chord is an approximation that curved paths and branches can violate; the native-vessel experiment in Sec. 5.5 evaluates this case directly.

3.3. Forecast-Conditioned Readouts

A readout tests, at every candidate location in the core, whether the crop contains a compact bright signal beyond what the forecast explains. We first fix the local quantities it operates on.

Local windows and projections. For a search center x in the core, let $W_x(p) = \exp(-\|p - x\|^2/2\sigma_W^2)$ for voxels p with $\|p - x\| \le r_W$ and $W_x(p) = 0$ otherwise, where σ_W sets the width of the window and r_W its support. These weights define a local inner product and norm,

$$\langle a, b\rangle_W = \sum_p W_x(p)\, a(p)\, b(p), \qquad \|a\|_W^2 = \langle a, a\rangle_W, \tag{2}$$

for any two volumes a, b. Let B be the matrix whose four columns are a constant and the three spatial coordinates of the window, and let

$$P_x a = a - B\,(B^\top W_x B)^{-1} B^\top W_x\, a \tag{3}$$

be the W_x-weighted projection onto the orthogonal complement of B, which removes the local background level and any linear intensity drift. A lesion of radius r is represented by the compact template $T_{x,r}(p) = [1 - \|p - x\|^2/r^2]_+^{0.8}$, where $[\cdot]_+ = \max(\cdot, 0)$ and the radius r ranges over a small set $\mathcal{R}$ of lesion sizes. Applying the projection to the observation, the forecast and the template yields the three local quantities

$$z = P_x y, \qquad u = P_x f, \qquad t = P_x T_{x,r}, \tag{4}$$

which we model as

$$z \approx g\,u + h\,t, \qquad h \ge 0. \tag{5}$$

Here the vessel gain g scales the forecast and the lesion amplitude h scales the template; h is constrained to be non-negative because lesions are brighter than the parenchyma.

The no-lesion fit. The exterior fit already estimates the vessel amplitude, which corresponds to $g = 1$. Under the null hypothesis $h = 0$ we therefore allow g to deviate from 1 only at a cost. With a penalty weight $\rho > 0$ and $\lambda = \rho\|u\|_W^2$, which makes the penalty scale with the forecast energy, the null loss is

$$J_0 = \min_g \; \|z - gu\|_W^2 + \lambda(g - 1)^2. \tag{6}$$

Setting the derivative with respect to g to zero gives the closed-form minimizer

$$g_0 = \frac{\langle u, z\rangle_W + \lambda}{G}, \qquad G = \|u\|_W^2 + \lambda, \tag{7}$$

which interpolates between the least-squares gain $\langle u, z\rangle_W / \|u\|_W^2$ (small ρ) and the exterior gain 1 (large ρ).

Adding the lesion term. For any fixed treatment of the gain, adding the term $h\,t$ is a one-dimensional nonnegative least-squares problem, whose solution has a simple closed form.

Lemma 1. *Let $J(h) = J_0 - 2ah + h^2 d$ with $d > 0$. Then $\min_{h\geq 0} J(h)$ is attained at $h^\star = [a]_+/d$, and the loss decreases by*

$$\Delta J = J_0 - J(h^\star) = \frac{[a]_+^2}{d}. \tag{8}$$

Proof. J is a convex quadratic with unconstrained minimizer a/d. If $a > 0$ this point is feasible and $J_0 - J(a/d) = a^2/d$; if $a \leq 0$, J is nondecreasing on $h \geq 0$ and the minimum is J_0 at $h = 0$. □

The correlation a measures how well the template matches the part of z not explained by the vessel, and d is the effective energy of the template. The four readouts differ only in how they treat the gain and therefore in a and d:

$$\begin{aligned}
&\text{CT-only:} && a = \langle t, z\rangle_W, && d = \|t\|_W^2,\\
&\text{Fixed:} && a = \langle t, z-u\rangle_W, && d = \|t\|_W^2,\\
&\text{Plug-in:} && a = \langle t, z\rangle_W - g_0\langle t, u\rangle_W, && d = \|t\|_W^2,\\
&\text{Profile:} && a = \langle t, z\rangle_W - g_0\langle t, u\rangle_W, && d = d_{\text{prof}}.
\end{aligned} \tag{9}$$

The CT-only readout ignores the forecast ($u = 0$). Fixed subtraction keeps the exterior gain $g = 1$, so its null loss is $J_0 = \|z-u\|_W^2$. The plug-in readout holds the null estimate g_0 fixed. The joint profile (profile for short) instead fits the gain and the lesion amplitude together; the following result shows that this again has the form of Lemma 1.

Proposition 1. *Let $J(g,h) = \|z - gu - ht\|_W^2 + \lambda(g-1)^2$. Then $\min_g J(g,h)$ is attained at $g(h) = g_0 - h\langle u, t\rangle_W/G$, and*

$$\begin{aligned}
\min_g J(g,h) &= J_0 - 2ah + h^2 d_{\text{prof}},\\
d_{\text{prof}} &= \|t\|_W^2 - \langle t, u\rangle_W^2/G,
\end{aligned} \tag{10}$$

with $a = \langle t, z\rangle_W - g_0\langle t, u\rangle_W$ as for the plug-in readout.

Proof. For fixed h, J is the null objective of Eq. (6) applied to $w = z - ht$, so its minimum is $\|w\|_W^2 + \lambda - (\langle u, w\rangle_W + \lambda)^2/G$ at $g = (\langle u, w\rangle_W + \lambda)/G = g(h)$. Expanding $\|w\|_W^2 = \|z\|_W^2 - 2h\langle t, z\rangle_W + h^2\|t\|_W^2$ and $\langle u, w\rangle_W + \lambda = Gg_0 - h\langle u, t\rangle_W$, and subtracting the same expression at $h = 0$, which equals J_0, gives $-2h(\langle t, z\rangle_W - g_0\langle u, t\rangle_W) + h^2(\|t\|_W^2 - \langle u, t\rangle_W^2/G)$. □

Profile and plug-in therefore coincide when template and forecast are orthogonal after projection; where they overlap, $d_{\text{prof}} < \|t\|_W^2$ and profiling assigns more weight to the same residual.

Why the gain treatment matters at the lesion. The difference between the readouts is sharpest when a lesion is present.

Proposition 2. *Let $z = z_0 + h_\star t$ with $h_\star > 0$, and let a_{fix} and a_{plug} denote the correlations of the fixed and plug-in readouts in Eq. (9). Then, relative to z_0,*

$$\Delta a_{\text{fix}} = h_\star\|t\|_W^2, \qquad \Delta a_{\text{plug}} = h_\star\Big(\|t\|_W^2 - \frac{\langle u, t\rangle_W^2}{G}\Big). \tag{11}$$

Proof. a_{fix} is linear in z with coefficient t, which gives the first identity. For the plug-in readout, g_0 is itself linear in z and increases by $h_\star\langle u, t\rangle_W/G$, which lowers $-g_0\langle t, u\rangle_W$ by $h_\star\langle u, t\rangle_W^2/G$. □

A gain estimated inside the core can thus absorb part of the lesion whenever forecast and template overlap, whereas a fixed gain cannot absorb it but inherits any amplitude error of the exterior fit. For an empty forecast ($u = 0$) all four readouts reduce to the same local test, so differences on such crops arise only from the fitted classifier coefficients.

3.4. Scoring, Search and Classification

Loss reductions at different locations are not directly comparable, because the residual energy varies with local texture. We therefore normalize the square root of Eq. (8) by the root-mean-square null residual of the window, $\sqrt{J_0/\sum_p W_x(p)}$, combined with a floor $\tau > 0$ that prevents division by near-zero residuals in very smooth regions. At each center x and radius r this gives the normalized score

$$m_r(x) = \frac{[a]_+}{\sqrt{d}\,\sigma_0}, \qquad \sigma_0 = \sqrt{\frac{J_0}{\sum_p W_x(p)} + \tau^2}. \tag{12}$$

The per-radius maxima $\max_x m_r(x)$ are passed to a ridge-logistic classifier, and the overall maximizer over x and r defines the location $L(y)$ (Algorithm 1). All classifiers share 12 local intensity and Hessian summaries, three template maxima from a larger window and three CT-only maxima. The CT + shell control adds its four shell summaries, and each forecast-conditioned readout adds its three conditioned maxima, so that fixed subtraction, plug-in and profile have identical feature counts and regularization budgets. Each readout is tuned over three coupled settings of the gain penalty ρ, the floor τ and the ridge strength (Sec. S2).

Implementation details. The forecaster samples the masked, smoothed crop at 2,400 nearly uniform anchors on the sphere of radius $R_a = 15.5\,\text{mm}$, tests continuation at $R_1 = 17.8\,\text{mm}$ and $R_2 = 20.4\,\text{mm}$, and tapers the field between $R_t = 22\,\text{mm}$ and $26\,\text{mm}$ with a cubic smoothstep; the remaining forecaster constants are listed in Sec. S1. The

Algorithm 1 Forecast-conditioned candidate scoring

Require: crop y, center c, readout $\in$ {CT-only, fixed, plug-in, profile}
$f \leftarrow F(y|_E)$ ▷ forecast from voxels beyond 14 mm
for centers x in the core, radii $r \in \mathcal{R}$ **do**
  $z, u, t \leftarrow P_x y,\ P_x f,\ P_x T_{x,r}$ ▷ affine residuals
  $J_0, g_0 \leftarrow$ no-lesion fit, Eq. (6) (fixed: $g = 1$)
  $a, d \leftarrow$ correlation and energy for the readout
  $m_r(x) \leftarrow [a]_+ / \left(\sqrt{d}\,\sigma_0\right)$ ▷ Eq. (12)
end for
$\mathbf{q} \leftarrow$ shared CT features $\oplus\, (\max_x m_r(x))_r$
$A(y) \leftarrow [\mathrm{logit}(\mathbf{q}) > \kappa]$ ▷ κ at 10% FPR
$L(y) \leftarrow \arg\max_{x,r} m_r(x)$
return $A(y)$, $L(y)$

readout windows use $\sigma_W = 3.4\,\mathrm{mm}$ and $r_W = 6\,\mathrm{mm}$, and the template radii are $\mathcal{R} = \{1.85, 2.7, 3.7\}\,\mathrm{mm}$. The gain penalty ρ, the noise floor τ and the ridge strength are tuned jointly over three coupled settings listed in Sec. S2.

4. Experiments

4.1. Datasets

We evaluate on three settings derived from LUNA16 [2, 15] and HiPaS [4], whose fitting, development, calibration and test roles are listed in Table S1 of the supplement. In each setting a *block* is a patient (LUNA16) or a scan (HiPaS), and every control crop is paired with a bank of 18 insertions that vary lesion shape, contrast and offset.

The *low-texture LUNA16 subset* comprises 549 neighborhoods from 183 patients, each with a single, crossing or skew added vessel. Its core intensity standard deviation is capped at 42 HU to isolate cases in which vessels compete strongly with the lesion, and its lesions are compact, radially symmetric signals. Thresholds and the analysis for this subset were fixed before testing, but because these cases also informed the design of the shared readouts, we treat readout comparisons on them as development results.

The *broader LUNA16 held-out set* is the main test set. It uses 170 patients disjoint from all fitting, development and calibration data, with four parenchymal neighborhoods per patient (680 controls and 12,240 insertions). It imposes no texture cap (median core SD 117 HU), uses added vessels that bend and taper, and introduces bilobed and part-solid-like lesions at three contrasts (84, 196 and 364 HU) and three offsets (0, 4 and 7 mm) from the proposal center. These shapes differ from the search templates, so the test does not reward template matching; the sampling constraints and the lesion model are detailed in Sec. S3.

Finally, the *HiPaS native-vessel set* tests whether the effect survives real anatomy. It consists of 72 noncontrast scans from a different source with manual artery and vein annotations. Six neighborhoods per scan are centered between nearby annotated arteries and veins, yielding 432 controls and 7,776 insertions without any added vessel (median core SD 253 HU). The annotations define only the sampling population and the analysis strata; the scorer observes the image alone.

4.2. Experimental Settings

Classifiers for the broader set are fitted on 80 patients and tuned on 85 others over the three coupled settings of (ρ, τ) and the ridge strength (Sec. S2). Before the test images were rendered, we fixed the cohort, the insertion bank, the analysis plan and three scorers selected on development: the aligned profile as the *lead* scorer, fixed subtraction as the strongest alternative readout, and the rotated profile with a refitted classifier. The primary criterion was a crop-sensitivity gain of at least 5 pp at 10% FPR, with a positive lower confidence limit against both the rotation and fixed subtraction; localized sensitivity was a prespecified secondary endpoint. A development bootstrap indicated that approximately 123 patients provide 80% power to detect a true 5-pp gain over the rotation, and 74 patients over fixed subtraction, so the 170 test patients are sufficient for both comparisons.

4.3. Baselines

All readouts of Sec. 3.3 are evaluated with both placements, together with three reference scorers. The *CT-only* scorer uses the shared features and the CT-only readout, and the *CT + shell* scorer adds the four shell summaries, which supply the exterior information without a spatial field. The *Hessian* baseline subtracts the same forecast at a strength of 0.35, 0.7 or 1.0 before computing a standard compact-curvature statistic at Gaussian scales of 1.3, 2.5 and 4.0 mm; it measures the contribution of a different readout when the nuisance estimate is held fixed. The learned baseline is a hybrid CNN with a small 3D feature extractor and a shared CT feature path, whose second input is either smoothed CT (the *CT-only CNN*), the aligned forecast, the rotated forecast or the residual $y - f$; architecture, data, seeds and training budget are identical across inputs (Sec. S6). Apart from the detector-score analysis on LNDb in Sec. 5.6, we do not compare against complete nodule-detection systems: they couple proposal generation, context and classification, so a placement sham cannot be applied to their vessel evidence alone, which is the quantity under test.

4.4. Evaluation Metrics

We report two operating points. The *matched* operating point sets each scorer to a 10% candidate false-positive rate (FPR) on the evaluation controls via the order statistic of their scores, and therefore measures ranking at a fixed budget. The *frozen* operating point applies cutoffs derived from 700 independent negative crops of 175 LUNA16 calibration patients, accepting scores strictly above the 10% order statis-

Readout	Low-text. sens.	Broader held-out set sens.	loc. sens.	frozen
Fixed, aligned	**50.75**	**24.05**	**15.38**	**24.74 / 10.59**
Fixed, rotated	36.03	20.81	5.72	21.18 / 10.15
Plug-in, aligned	39.45	22.54	11.54	23.86 / 10.44
Plug-in, rotated	37.39	21.26	6.92	19.81 / 9.12
Profile, aligned	46.50	23.32	13.45	23.79 / 10.59
Profile, rotated	37.13	21.23	7.35	19.29 / 8.82
Prof. rot., aligned clf.	–	19.26	9.19	19.20 / 10.00
CT + shell	36.60	21.50	13.49	21.66 / 10.29
CT only	32.88	21.39	13.37	23.78 / 11.62
Hessian subtr.	47.34	14.31	–	14.31 / 10.29
Lead minus … (pp, 95% CI)				
− refitted rotation	–	+2.1 [−0.3, +4.4]	+6.1 [+4.4, +7.8]	–
− fixed subtr.	–	−0.7 [−3.0, +0.5]	−1.9 [−3.3, −1.0]	–

Table 1. Sensitivity (%) at the matched 10% FPR on the low-texture subset and the broader held-out set, and at the frozen LUNA16 cutoffs (sensitivity / achieved FPR). Bold: best aligned readout per column. A dash marks a quantity not computed in that setting.

tic; its achieved FPR may differ between scorers. At either operating point, *crop sensitivity* is the fraction of lesion crops accepted, and *localized sensitivity* additionally requires the returned location to lie within 3 mm of the insertion center $x_\star$,

$$\Pr\big(A(y)=1,\ \|L(y)-x_\star\| \le 3\,\mathrm{mm} \mid Y=1\big). \quad (13)$$

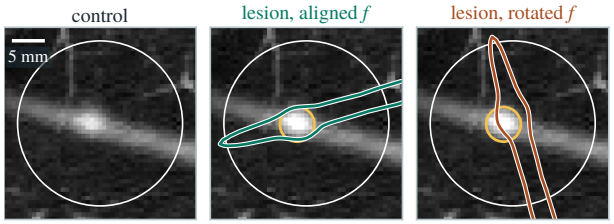
Controls count against the FPR budget irrespective of their returned location.

Differences between two scorers are reported as "$+x$ pp $[a, b]$": x is the paired difference in sensitivity in absolute percentage points (pp), and $[a, b]$ its 95% confidence interval from 3,000 bootstrap resamples of blocks, stratified by release fold. Each resample retains a block's full insertion bank and controls, so both scorers are always evaluated on the same crops. For matched intervals the cutoff is recomputed in every draw. For frozen cutoffs we report intervals conditional on the calibration set and, separately, intervals that also resample the calibration patients.

5. Results

Our experiments address three questions that follow from the toy example in Sec. 1: whether a correctly placed forecast changes the scorer's decisions on held-out data, whether it changes the crop decision or the returned location, and which readout and which insertion conditions realize the effect. We then test whether the effect survives the move to native vessels and to thresholds calibrated on another acquisition.

(a) Same exterior, aligned vs rotated forecast



(b) Paired effect of alignment (profile readout, 10% FPR)

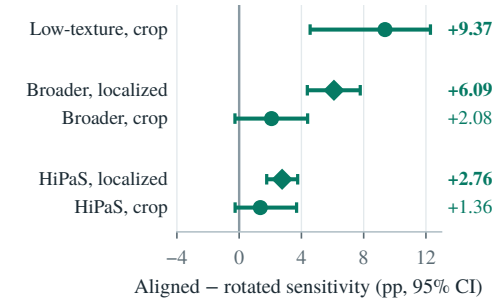


Figure 2. The same forecast helps only when it is correctly placed. (a) Development example with the aligned and rotated forecast outlines; gold ring: inserted lesion; thin circle: 14-mm mask. (b) Aligned minus rotated sensitivity of the profile readout at the matched 10% FPR; circles: crop, diamonds: localized sensitivity; bold values exclude zero.

5.1. Effect of Forecast Alignment

We first ask whether the aligned forecast improves the prespecified scorer on the broader held-out set. Fig. 2 summarizes the paired effect of alignment in all three settings, and Table 1 lists every scorer. The broader set is considerably harder than the development data: at the matched 10% FPR the readouts accept between about 19% and 24% of lesion crops, and the Hessian baseline only 14.31%. The aligned profile increased crop sensitivity by $+2.08$ pp [-0.27,+4.40] over the rotation with a refitted classifier and by $+4.05$ pp [+1.37,+6.23] over the rotation evaluated with the aligned classifier. At frozen cutoffs the difference was $+4.50$ pp, albeit at achieved FPRs of 10.59% *vs.* 8.82%; including calibration uncertainty, the interval was [-0.59,+8.31] pp and the upper limit on excess FPR 5.00 pp. The prespecified crop-level criterion was therefore not met. The two controls without a spatial field behave as expected: the CT-only and CT + shell scorers accept 21.39% and 21.50% of lesion crops, so exterior summaries alone add essentially nothing. Localization, in contrast, improved substantially (Fig. 3): the rotated profile returned the lesion location for 7.35% of insertions and the aligned profile for 13.45%, a gain of $+6.09$ pp [+4.38,+7.78]. The same pattern holds for every forecast-conditioned readout: with the rotated forecast, fixed subtraction localizes 5.72%, the plug-in readout

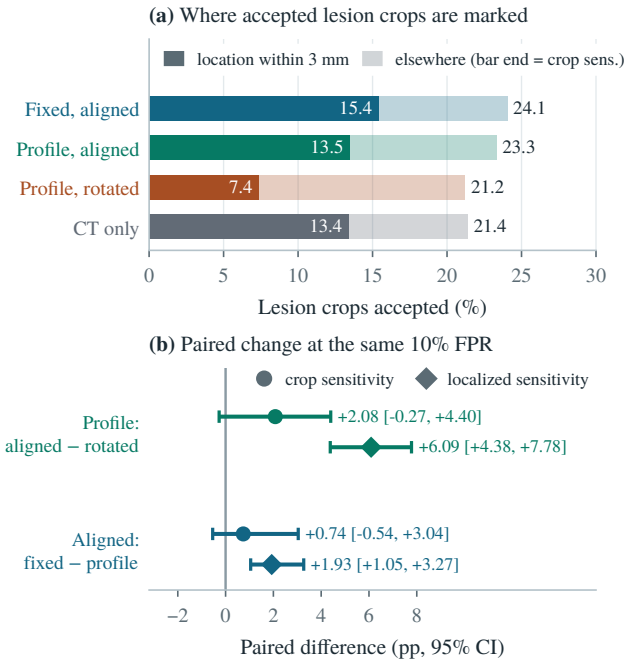


Figure 3. Acceptance and localization are distinct decisions (broader held-out set, matched 10% FPR). (a) Accepted lesion crops whose returned location lies within 3 mm of the lesion (dark) or elsewhere (light). (b) Paired differences with 95% CIs.

6.92% and the profile 7.35% of insertions, all below the CT-only scorer (13.37%), whereas with the aligned forecast they reach 15.38%, 11.54% and 13.45%. A misplaced forecast thus localizes worse than no forecast at all, consistent with the toy example: its subtraction removes nothing from the vessel and distorts the background elsewhere.

The low-texture subset, where vessels compete most strongly with the lesion, shows the placement effect already at the crop level. There the aligned profile outperformed its rotation by $+9.66\,\mathrm{pp}$ [+8.10,+11.19] at frozen cutoffs (achieved FPR 10.38% *vs.* 10.02%) and by $+9.37\,\mathrm{pp}$ [+4.56,+12.29] at matched FPR (Fig. 2b). Under the shared readouts, fixed subtraction gained $+14.72\,\mathrm{pp}$ [+11.11,+17.05] over its rotation and $+14.15\,\mathrm{pp}$ [+10.73,+16.44] over the CT + shell control, gaining 2,001 and losing 546 positive decisions relative to the rotation (Table 1). Because these cases informed the design of the readouts, we regard them as supporting rather than confirmatory evidence.

5.2. Acceptance versus Localization

A crop-by-crop comparison of the two placements explains the difference between the endpoints. Of the net gain of 746 localized insertions, about a third arises from crops accepted under both placements, where alignment moves the returned location onto the lesion (386 crops, against 124 moved away); the remainder comes from crops that only the aligned profile accepts (647 *vs.* 163; Table S3). Localization therefore improves considerably more than acceptance. This behavior follows from the competition illustrated in Fig. 1: if a vessel response $m(x_v)$ exceeds a lesion response $m(x_\ell)$ and conditioning lowers them by d_v and d_ℓ, the lesion becomes the maximum when $d_v - d_\ell > m(x_v) - m(x_\ell)$, regardless of whether the crop decision changes.

The localization gain does not depend on the choice of readout or on refitting. In a post hoc analysis, fixed subtraction compared with its own separately tuned rotation gains $+9.66\,\mathrm{pp}$ [+8.24,+11.38] in localized sensitivity on the broader set and $+2.39\,\mathrm{pp}$ [+1.22,+3.28] on HiPaS. Rotating only the field under the aligned classifier yields $+4.26\,\mathrm{pp}$ [+2.39,+5.83] and $+0.51\,\mathrm{pp}$ [-0.15,+1.09], respectively.

5.3. Readout Flexibility and Learned Baselines

Additional flexibility in the vessel amplitude did not improve performance (Table 1). On the broader set, the plug-in readout, which re-estimates the gain inside the window without a lesion term, is the weakest aligned readout in both crop (22.54%) and localized sensitivity (11.54%). On the low-texture subset, the profile trailed the Hessian baseline by $0.84\,\mathrm{pp}$ [-6.15,+2.43], whereas fixed subtraction exceeded it by $3.41\,\mathrm{pp}$ [-0.23,+6.05]. On the broader set, the profile trailed fixed subtraction by $0.74\,\mathrm{pp}$ [-3.04,+0.54] in crop sensitivity, and fixed subtraction localized $1.93\,\mathrm{pp}$ [+1.05,+3.27] more insertions (15.38% of all insertions). On 18 development backgrounds with exact fields, where the forecast equals the added vessel, fixed subtraction and profiling both ordered 67.6% of insertion/control pairs correctly, compared with 62.0% for the plug-in readout. These results are consistent with the analysis in Sec. 3.3: a gain estimated inside the core can absorb part of the lesion.

The CT-only CNN reached 67.54% development AUROC and 36.28% sensitivity at the 10% cap, comparable to the ridge classifier on the CT-only readout (68.08% and 36.47%). Table 2 compares its four second inputs on the low-texture subset: smoothed CT reaches 34.28% sensitivity, the aligned and rotated forecasts 35.14% and 35.68%, and the residual $y - f$ 36.44%. The aligned-minus-rotated difference was $-0.54\,\mathrm{pp}$ [-2.80,+0.62], and the residual input changed sensitivity by $+2.16\,\mathrm{pp}$ [-1.19,+4.44]. The network is not under-trained: it fits small training sets almost perfectly, and its development AUROC increases with the number of fitting patients (Sec. S6 and Fig. S1). A classifier that pools learned features can therefore use the same field without exploiting its placement; the placement gain is specific to a readout in which the field can suppress a competing peak.

5.4. Analysis by Insertion Condition

Fig. 4 decomposes the paired crop-level gain by insertion condition, evaluated at each setting's global cutoff. Gains

Second input	Epoch	Dev. AUROC	AUROC	Sens.
CT only (smoothed CT)	30	67.54	68.78	34.28
Aligned forecast	20	67.54	68.81	35.14
Rotated forecast	30	67.81	69.13	35.68
Residual $y - f$	30	68.16	69.46	36.44

Table 2. Hybrid CNN with four second inputs on the low-texture subset (%), with identical architecture, data, seeds and budget. The ridge classifier on the CT-only readout reaches 69.17% AUROC and 32.88% sensitivity.

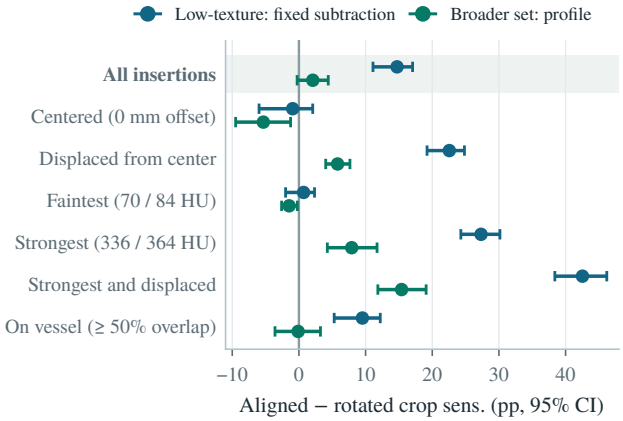


Figure 4. Placement gains by insertion condition: aligned minus rotated crop sensitivity at each setting's 10% FPR cap, for fixed subtraction on the low-texture subset and the profile on the broader set. Rows are not mutually exclusive.

concentrate where a vessel is most likely to produce the search maximum, namely for displaced, high-contrast insertions, which gained $+15.40$ pp [+11.84,+19.08] on the broader set. Centered insertions changed by -5.34 pp [-9.49,-1.25] and the faintest insertions by -1.47 pp [-2.60,-0.24], while insertions overlapping a vessel by at least 50% were unaffected (-0.12 pp [-3.59,+3.23]). A centered lesion is often already the strongest response, and a faint lesion leaves a weak residual even when the vessel is well explained. The low-texture subset exhibits the same pattern with larger effects.

We also evaluated a forecast-dominance score that is available before scoring, defined as the peak forecast relative to the texture of the shell (Sec. S5). Crops in the upper half of this score, 40.7% of the broader set, show a gain of $+3.29$ pp [+0.04,+6.03], but the high-minus-low interaction of $+2.03$ pp [-1.62,+5.11] is not resolved. Within the high-dominance band, alignment also raises the frozen FPR (11.19% *vs*. 9.39%; Sec. S5), so the score describes where alignment matters but does not by itself identify a better operating point.

	Matched 10% FPR		LUNA16 cutoffs
Readout	sens.	loc. sens.	sens. / FPR
Fixed, aligned	16.26	5.66	1.93 / 0.46
Fixed, rotated	14.30	3.27	5.00 / 2.55
Plug-in, aligned	15.64	5.31	2.31 / 0.69
Plug-in, rotated	14.63	2.84	6.53 / 3.94
Profile, aligned	15.52	5.47	2.15 / 0.69
Profile, rotated	14.16	2.70	6.62 / 4.63
Profile, rot., aligned clf.	14.97	4.95	0.91 / 0.23
CT + shell	14.45	4.21	4.69 / 2.55
CT only	14.47	3.38	10.83 / 7.87
Hessian subtraction	11.33	–	12.04 / 10.88

Table 3. Sensitivity (%) on HiPaS native vessels (432 controls, 7,776 insertions) at the within-HiPaS 10% FPR cap and at the frozen LUNA16 cutoffs.

5.5. Transfer to Native Vessels

On the 72 HiPaS scans, 7,713 of the 7,776 insertions lie within 2 mm of an annotated artery or vein. At matched FPR, alignment improved localized sensitivity by $+2.76$ pp [+1.77,+3.76] and crop sensitivity by $+1.36$ pp [-0.26,+3.69]. The localization gain therefore persists when real anatomy, including curved vessels that a straight chord approximates poorly, supplies the competing response. Table 3 lists all scorers. Fixed subtraction (5.66%), the profile (5.47%) and the plug-in readout (5.31%) localize roughly twice as many insertions as their rotations (3.27%, 2.70% and 2.84%), and more than the CT-only (3.38%) and CT + shell (4.21%) scorers; alignment produced 284 newly localized insertions against 69 lost. At the crop level, all scorers accept 14–16% of lesion crops except the Hessian baseline (11.33%).

5.6. Ranking versus Calibration

When applied unchanged to HiPaS, the LUNA16 cutoffs yielded 2.15% sensitivity at 0.69% FPR for the aligned profile, compared with 6.62% at 4.63% for its rotation (Table 3 and Fig. 5b). Most scorers accepted far fewer HiPaS controls than the nominal 10%, so their realized operating points shifted. Fig. 5 shows the ROC curves underlying these operating points, computed with the negatives of each cohort. On the broader set, both aligned readouts lie above the rotation and the CT-only scorer between approximately 3% and 25% FPR, so the matched result does not depend on the choice of a 10% cap. On HiPaS the crop-level curves nearly coincide, and the gain is confined to localization. The frozen LUNA16 cutoffs (open circles) fall close to the 10% cap on LUNA16 but far below it on HiPaS. Table 4 quantifies the contribution of calibration uncertainty: resampling the 175 calibration patients widens the broader lead-minus-rotation interval from $[+2.62, +6.32]$ to $[-0.59, +8.31]$ pp. On HiPaS the lead scorer accepts fewer insertions and fewer controls than its rotation, which reflects a shift of the operating point rather

Contrast (interval type)	Sensitivity Δ	FPR Δ
Broader LUNA16		
Lead−rot. (frozen)	+4.50 [+2.62, +6.32]	+1.76 [−0.59, +4.12]
Lead−rot. (recal.)	+4.50 [−0.59, +8.31]	+1.76 [−2.35, +5.00]
Lead−fixed (frozen)	−0.95 [−1.85, −0.10]	+0.00 [−1.32, +1.32]
Lead−fixed (recal.)	−0.95 [−3.80, +1.32]	+0.00 [−2.21, +1.76]
HiPaS		
Lead−rot. (frozen)	−4.48 [−6.51, −2.58]	−3.94 [−6.02, −2.08]
Lead−rot. (recal.)	−4.48 [−7.09, −2.60]	−3.94 [−6.02, −1.62]
Lead−fixed (frozen)	+0.22 [−0.05, +0.62]	+0.23 [+0.00, +0.69]
Lead−fixed (recal.)	+0.22 [−0.32, +0.73]	+0.23 [+0.00, +0.69]

Table 4. Paired differences of the aligned profile at the frozen LUNA16 cutoffs (pp, 95% CI), with intervals conditional on the calibration set (frozen) or resampling it (recal.).

	Profile		Rotation		Hessian	
Condition	Sens.	FPR	Sens.	FPR	Sens.	FPR
Low-texture subset	46.85	10.38	37.19	10.02	38.39	5.46
Morphology shift	41.79	12.05	32.69	9.24	32.06	6.43
LNDb + morphology	28.04	6.49	29.27	9.14	36.09	12.09

Table 5. Frozen operating points (%) of the low-texture design on the subset itself and two transfer conditions, using its own 276 calibration negatives.

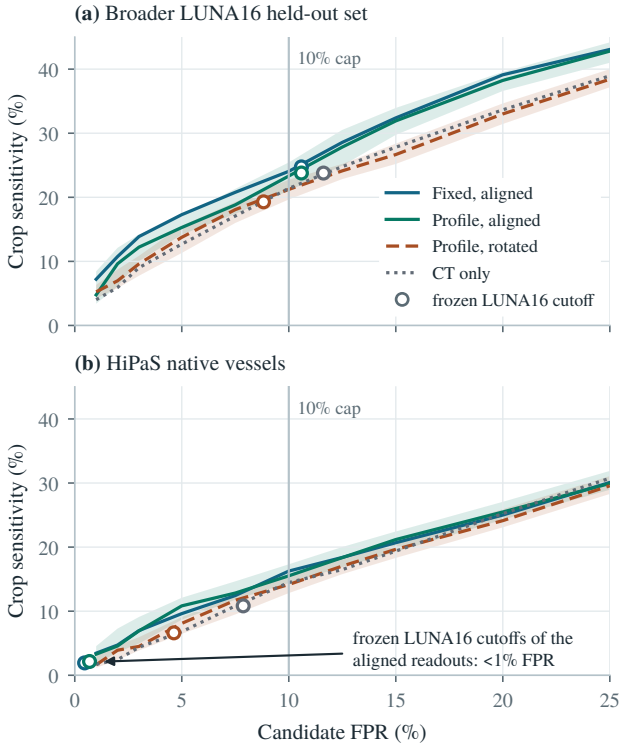


Figure 5. ROC curves with pointwise 95% bands for the profile placements; open circles mark the frozen LUNA16 cutoffs, which transfer poorly to HiPaS.

than a loss in ranking (Table 3).

Transfer conditions of the low-texture design. The same distinction appears in two additional transfer conditions of the low-texture design, a LUNA16 morphology shift and an independent LNDb acquisition with shifted insertions [10]. Neither met its full transfer criteria at frozen cutoffs (Table 5); on the low-texture subset itself, the upper limit on excess FPR relative to the CT-only scorer was 4.92 pp against a prespecified 4 pp. Within each test set, however, the profile and Hessian sensitivities at 10% FPR were 37.93% and 37.44% under the morphology shift and 33.15% and 32.94% on LNDb, so ranking transferred better than the cutoffs. On LNDb, adding the spatial features to the scores of a published detector changed mean FROC by 0.00 pp [-0.21,+0.23] relative to the full CT-only classifier and by −0.13 pp [-0.33,+0.35] relative to the sham.

6. Conclusion

We asked whether a vessel forecast helps a nodule candidate scorer because it is placed correctly or merely because it adds information. A paired design in which control and lesion crops share their exterior, together with a value-preserving rotation sham, answers this question directly: on held-out data, alignment changes the location the scorer returns far more than the decision it makes, and a plain fixed subtraction is enough to realize this benefit. The same localization gain appears on scans with expert-annotated native vessels, where real anatomy rather than simulated tubes supplies the competing response.

A vessel forecast can appear useful for several reasons, since its shape and amplitude correlate with the type of background in a crop; combining a value-preserving rotation sham, an exterior-summary control and shared readouts is what isolates the contribution of its *alignment*. Two practical implications follow. First, fixed subtraction combined with a compact-template search exploits the alignment as effectively as joint amplitude estimation, because a gain fitted inside the search region absorbs part of the lesion (Proposition 2), so it is a competitive baseline for more flexible nuisance models. Second, for pipelines that present a returned location to a reader, moving that location onto the lesion is valuable even when the number of accepted crops barely changes, whereas a gain in crop sensitivity can overstate the benefit when the returned location remains on the vessel.

The experiments also separate ranking from calibration. A matched-FPR effect establishes ranking separation at a given budget, whereas a shift in the negative score distribution changes the realized budget at an imported threshold. In our experiments, ranking gains from alignment transferred across acquisitions more reliably than calibrated cutoffs, so a

deployment-oriented claim would require negatives from the target acquisition. A reliability-based rule for using the forecast selectively did not outperform always using it (Sec. S6).

The study also has clear limits. All positive targets are inserted signals, so real nodules, scan-level proposal pipelines, partial-volume effects and per-scan false-positive rates remain untested. The prespecified crop-level criterion was not met, the localization robustness checks were conducted post hoc, and absolute localized sensitivity remains modest. Calibrated cutoffs, moreover, did not transfer across acquisitions even where ranking did.

These findings point to several directions. The most direct is a prespecified evaluation on adjudicated real nodules at the scan level with localized sensitivity as the primary endpoint. Because the learned CNN in our comparison used the forecast without exploiting its placement, a second direction is to build learned scorers that subtract or condition on an aligned forecast explicitly, which our results suggest is where the benefit originates. Replacing the straight-chord forecaster with one that follows curved and branching vessels should strengthen the effect on native anatomy, and calibrating thresholds with negatives from the target acquisition would turn the ranking gain into a transferable operating point. More broadly, a value-preserving placement sham is a simple and general control for any method that claims to benefit from spatial context.

LLM Usage Statement

OpenAI Codex assisted with study design, implementation, experiment execution, numerical analysis, literature lookup, figure preparation, and manuscript drafting and editing. Claude (Anthropic) assisted with revising the manuscript text and redrawing the figures from the recorded measurements. The illustration in Fig. 1a was created with BioRender. The quantitative figures were generated programmatically from recorded measurements. The qualitative panels use public CT crops or algorithmically generated phantoms. The authors are responsible for validating the work and its interpretation.

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